

Io's polar volcanic thermal emission indicative of magma ocean and shallow tidal heating models

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Supplementary Material to “Io’s polar volcanic thermal emission indicative of magma ocean and shallow tidal heating models”, by Ashley Gerard Davies¹, Jason Perry², David A. Williams³, David M. Nelson³, published in *Nature Astronomy*.

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Supplementary Table S1 to this letter consists of a csv file. For the 266 hot spots detected at 4.8- μm in JIRAM data from orbit PJ5 to orbit PJ43 inclusive (as shown in Figure 1), this file contains the following information:

Column A:	hot spot identifier (e.g., JRM001)
Column B:	name, if assigned to a specific feature or region
Column C:	latitude, °N
Column D:	longitude, °W
Column E:	JIRAM detections by orbit (i = inbound, o = outbound)
Column F:	maximum unsaturated 4.8- μm spectral radiance (GW/ μm).
Column G:	maximum 4.8- μm spectral radiance (GW/ μm) (includes saturated values)
Column H:	average 4.8- μm spectral radiance (GW/ μm) (excludes saturated values)
Column I:	minimum unsaturated 4.8- μm spectral radiance (GW/ μm)
Column J:	Earlier detections by asset (NIMS = <i>Galileo</i> Near Infrared Mapping Spectrometer; LEISA = <i>New Horizons</i> Linear Etalon Imaging Spectral Array).
Column K:	Notes

The following three figures are referred to in the Methods section of the paper.

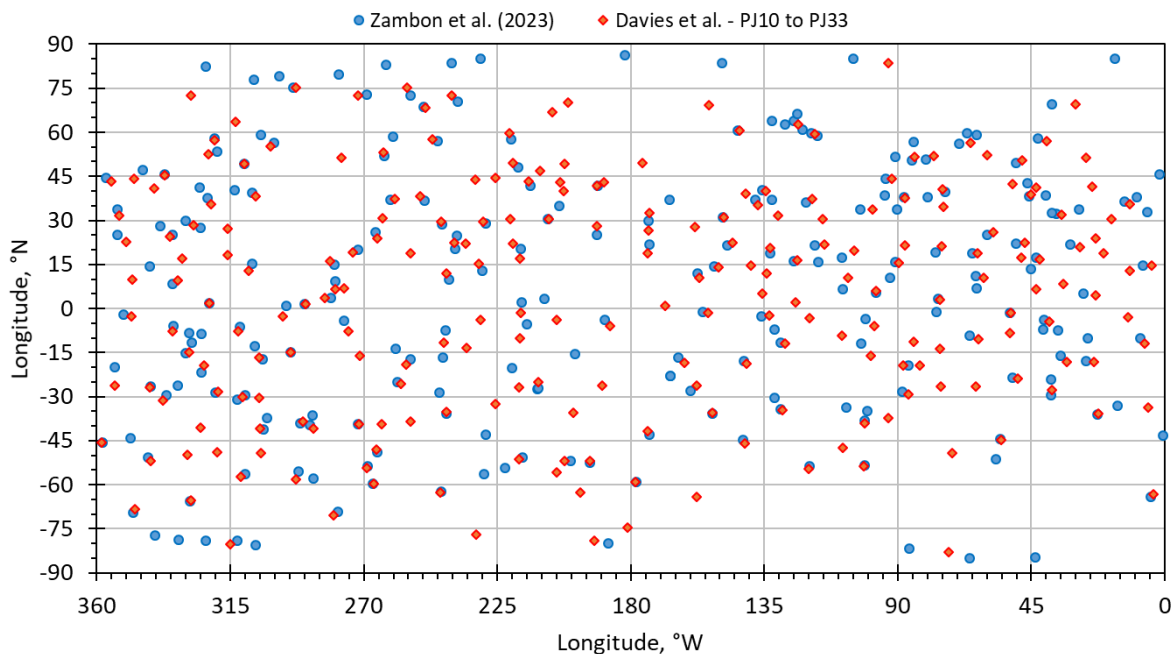


Figure S1. Comparison of hot spot detections in Zambon et al. (2023) and this analysis. This figure is limited to orbits PJ10 (December 2017) through PJ33 (April 2021) used by Zambon et al. We find fewer hot spots in polar regions, and more hot spots at lower latitudes. The discrepancies between analyses increase towards high latitudes, in particular above 75° .

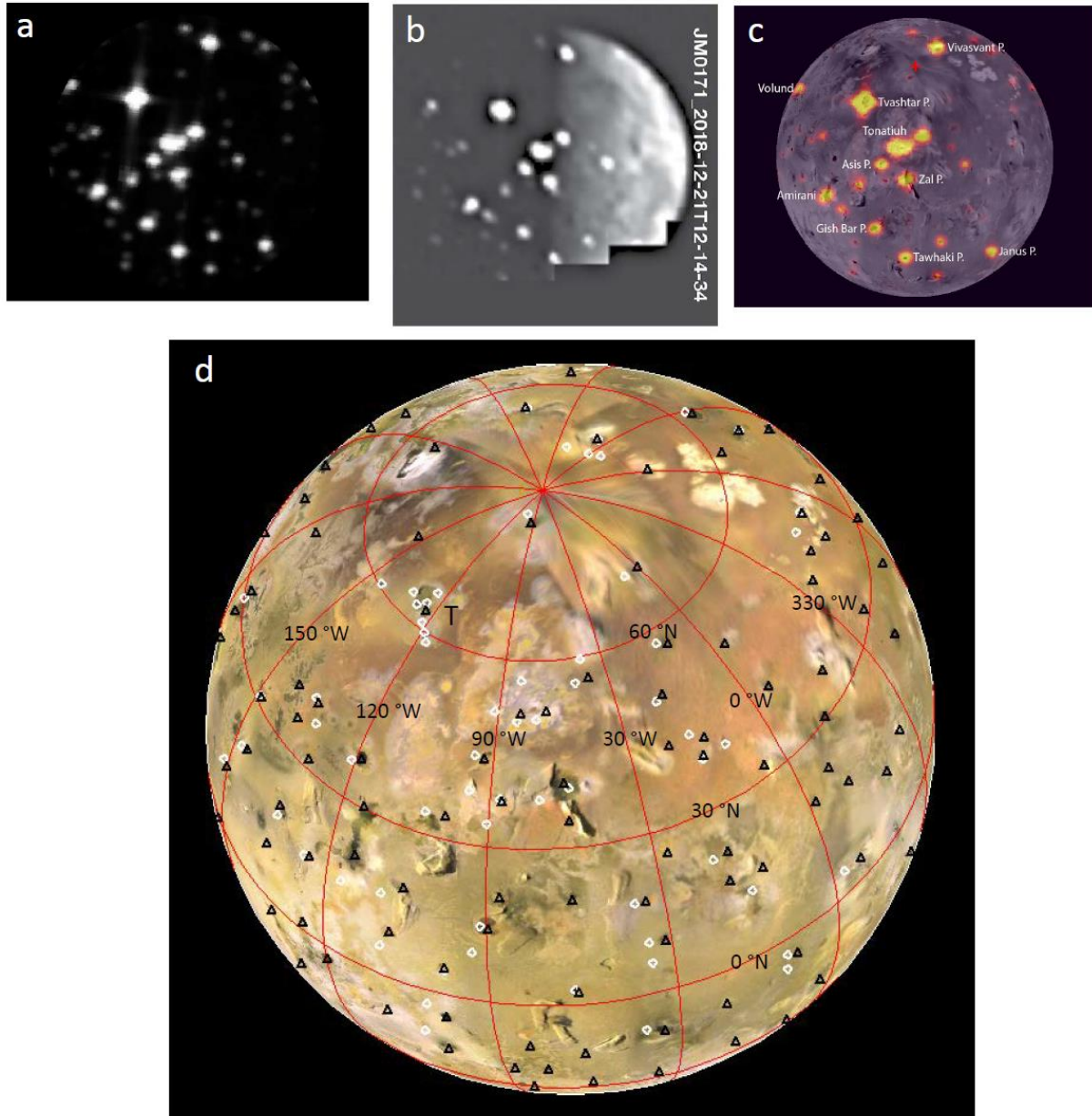


Figure S2. Comparison of superposition images for PJ17 (December 2018) from (a) this analysis and (b) Zambon et al. (2023). Main hot spots found in this analysis are shown in (c), where the north pole is marked with a red cross. Io in panels a-d is shown with the same viewing geometry and orientation. (d) shows hot spots identified in PJ17 data in this analysis (black triangles) compared with hot spots listed by Zambon et al. (white triangles) in PJ17 data. Data are plotted on the *Voyager-Galileo* mosaic²². We

identify 103 hot spots. Zambon et al. identify 64. At latitudes above 60 °N, we find 8 hot spots, but Zambon et al. find 13. At Tvashtar Paterae (a group of calderas centred on $\approx 119^\circ\text{W}$, $\approx 60^\circ\text{N}$, designated by “T” in panel d), we identify a single thermal source. Zambon et al. identify seven, with an eighth close by that we do not detect.

Saturated data. In JIRAM Reduced Data Record (RDR) products, band radiance values plateau at the point of detector saturation. Supplementary Figure S3 shows the saturation band radiance as a function of image integration time and includes the function relating exposure time to saturating band radiance. We identify and flag all saturated pixels and have created a saturation mask for every frame in every JIRAM observation of Io.

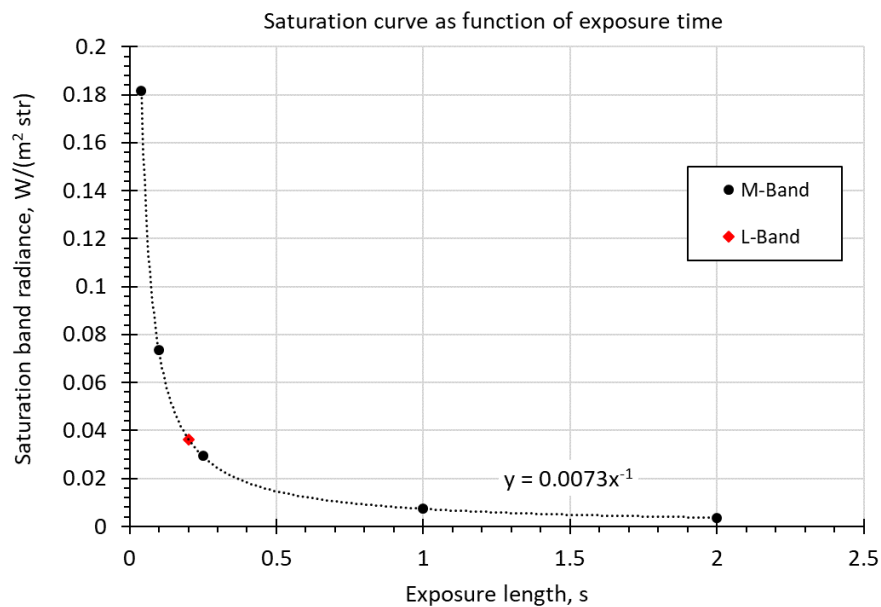


Figure S3. Graph showing the maximum band radiance from *Juno* JIRAM Reduced Data Record (RDR) images obtained from the NASA Planetary Data System as a function of observation integration time, for illustrative purposes. Note that these values are not used to determine the saturation threshold used in this analysis. Because of a non-linear detector response at high band radiance values (relating to high Data Number (DN) values in the Experiment Data Record [EDR]), we use a threshold of 5% below these values to flag saturated pixels. We also tested a recommended threshold 18% below the shown values (Mura et al., 2020) that flags some additional pixels as saturated, but we found that this makes no significant difference to our results. The 2 s data point is an extrapolation – JIRAM does not obtain exposures of this duration.